CSIT212
QUANTITATIVE METHODS

LESSON 1

Introduction to Quantitative Analysis and Data Analytics

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Learning Objectives

At the end of this lecture, students should be able to:

- Describe the quantitative analysis and its role in decision-making
- Distinguish quantitative and qualitative factors
- Explain the steps of the quantitative analysis approach
- Develop simple real-world quantitative models
- Conduct basic break-even analysis
- Recognize key issues and limitations of quantitative analysis

Lesson Outline

Part I
Overview of Quantitative Analysis

What is Quantitative Analysis?

Raw Data → Quantitative Analysis → Meaningful Information

Quantitative Analysis (QA) is a **scientific** and **systematic approach** to decision-making that uses:

- Mathematics
- Statistics
- Optimization techniques
- Data analysis

It **transforms raw data** into **meaningful information** for better decisions.

What is Quantitative Analysis?

Formal Definition

Quantitative Analysis is a **scientific approach** to **decision-making** in which **raw data** are **processed and manipulated** to produce **meaningful information** (Render, Stair, & Hanna, 2012).

Why Quantitative Analysis Matters

Decisions **based on intuition alone** can be **biased** or **inaccurate**.

Quantitative Analysis provides:

- Objective evaluation
- Measurable outcomes
- Repeatable and testable solutions

Quantitative Analysis is commonly used in:
IT systems planning, Business and management, Engineering, Healthcare, Government policy

Real-World Examples of Quantitative Analysis

CASE STUDY 1: In the mid 1990s, Taco Bell **saved over $150 million** using **forecasting** and **scheduling** quantitative analysis models.

CASE STUDY 2: NBC television **increased revenues by over $200 million** between 1996 and 2000 by using quantitative analysis to **develop better sales plans**.

CASE STUDY 3: **Continental Airlines saved over $40 million** in 2001 using quantitative analysis models to quickly **recover from weather delays** and **other disruptions**.

Quantitative vs Qualitative Data

Quantitative Data (Measurable) - expressed and interpreted numerically.

Qualitative Data (Non-measurable) - Difficult or impossible to express in numbers; usually expressed as categories

Important Note: **Good decisions** often require **BOTH** quantitative and qualitative considerations.

Quantitative vs Qualitative Data

TYPES OF DATA

Quantitative Data

- Investments
- Interest rates
- Inventory levels
- Demand
- Labor cost

Qualitative Data

- The weather
- State and federal legislation
- Technological breakthroughs

Part II
Data Analytics and Analytical Methods

What is Data Analytics?

Data Analytics process of examining **raw data** to uncover **patterns**, **trends**, **relationships**, and **useful insights** that can support decision-making.

It involves **collecting**, **cleaning**, **transforming**, and **analyzing** data using **statistical**, **mathematical**, and **computational techniques** to identify meaningful information and guide actions.

Data Analytics vs Data Analysis

Data Analysis focuses on **examining** and **interpreting data** to answer **specific questions**.

Data Analytics is a broader discipline that includes **data analysis plus the entire process** of turning data into decisions.

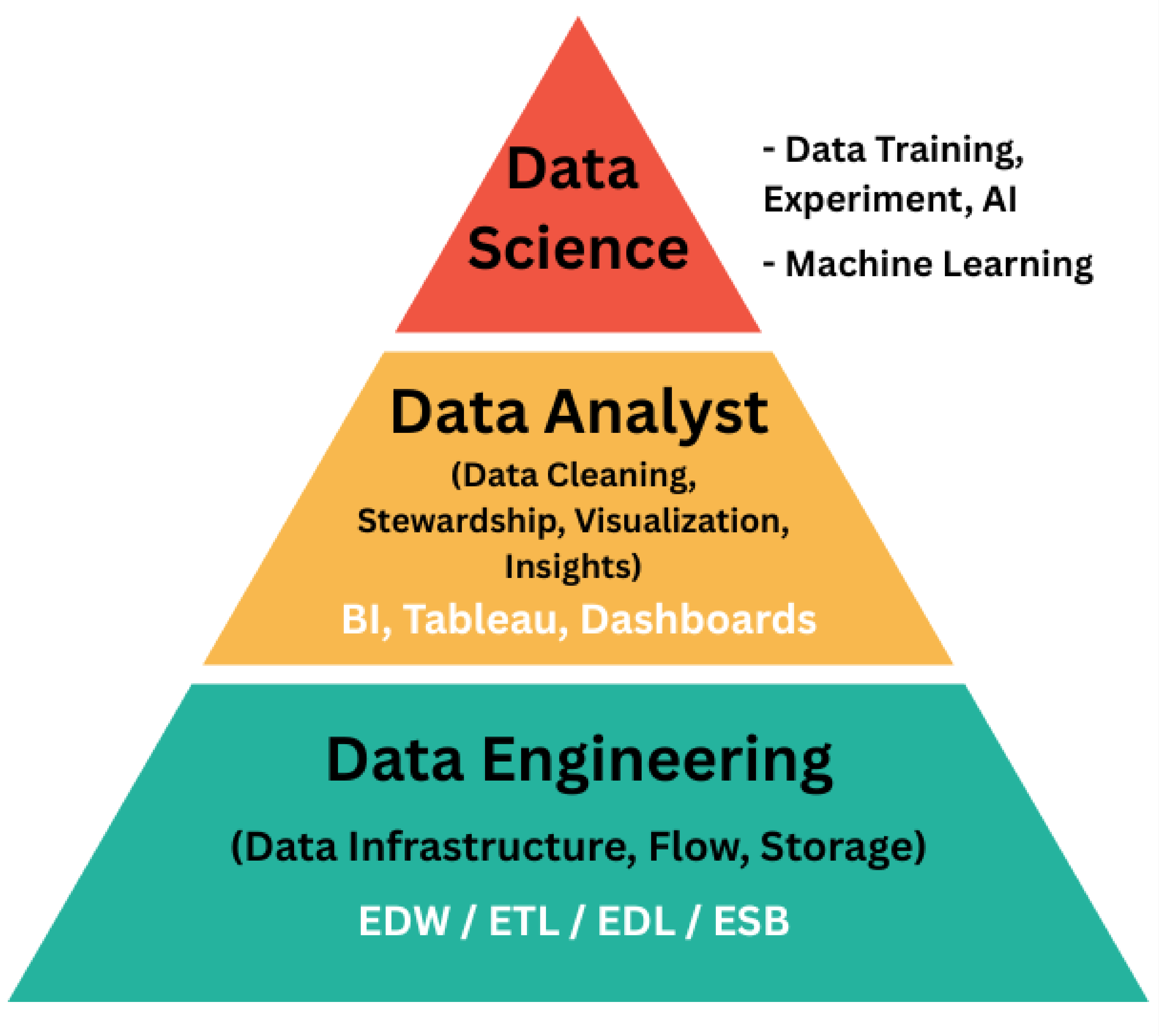
Importance of Data Analytics

Data analytics helps organizations to:

- Make **evidence-based decisions**
- Improve **efficiency** and **productivity**
- **Reduce risks** and **uncertainty**
- Gain **competitive advantage**
- **Predict future** outcomes

Data-driven/centered Careers

Data Science, **Data Analytics**, and **Data Engineering** are careers that have become increasingly important in today's data-driven world.



Development of Data Analytics

Early Data Processing Era (1960s–1980s)

Manual and **basic computer-based data handling**.

Characteristics:

- Data stored in **physical files** or early databases (**hierarchical, flat files**).
- Limited analytical power – mostly **reporting** and **simple statistics.**
- Batch processing, mainframe computers.

Technologies: COBOL, punch cards, early SQL databases.

Development of Data Analytics

Business Intelligence Era (1990s–2000s)

Systematic reporting and **dashboard creation** for business decisions.

Characteristics:

- Emergence of **relational databases** and data warehouses.
- **ETL (Extract, Transform, Load)** processes for structured data.
- Use of **OLAP (Online Analytical Processing)** for multi-dimensional analysis.

Technologies: Oracle, Microsoft SQL Server, SAP BusinessObjects.

Development of Data Analytics

Big Data & Advanced Analytics Era (2010s–Present)

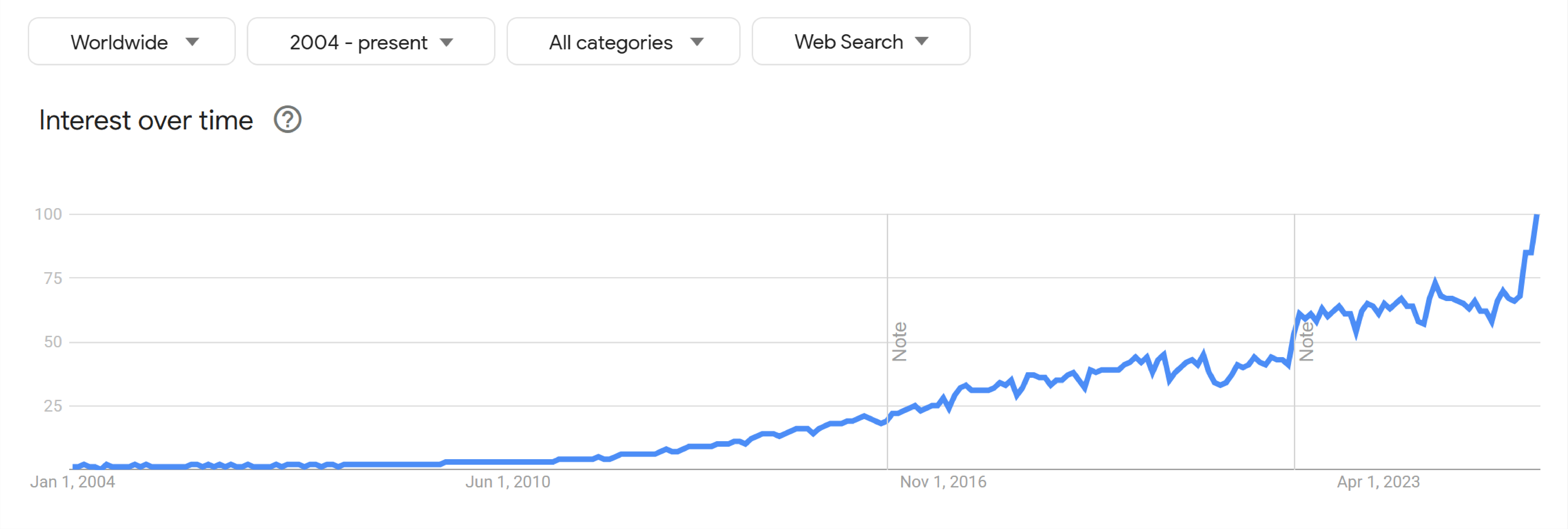
Real-time, **large-scale analytics** powered by AI and machine learning.

Characteristics:

- Explosion of unstructured/semi-structured data from IoT, social media, sensors.
- Use of distributed computing (Hadoop, Spark).
- Cloud-based analytics platforms and real-time streaming data analysis.

Technologies: Hadoop, Spark, AWS, Azure ML, TensorFlow.

Development of Data Analytics



Google Trends Graph of Searches on the term *Data Analytics (2004 – present)*

Categorization of Analytical Methods

(Types of Data Analytics)

Descriptive Analytics

- Answers "**What happened?**"
- Summarizes data to identify **patterns**, **trends**, and **anomalies.**
- Methods & Tools
  - Data Aggregation and Data Mining
  - Summary Statistics, Dashboards, and Reports

Example: Monthly sales **report** showing revenue trends.

Categorization of Analytical Methods

(Types of Data Analytics)

Diagnostic Analytics

- Answers "**Why did it happen?**"
- Explores historical data to find **causes** and **contributing factors** of events.
- Methods & Tools
  - Root Cause Analysis
  - Drill-Down and Data Discovery Techniques

Example: Identifying **reasons** for a sudden drop in website traffic.

Categorization of Analytical Methods

(Types of Data Analytics)

Predictive Analytics

- Answers "**What is likely to happen?**"
- Uses statistical models and machine learning to **forecast future** outcomes.
- Methods & Tools
  - Regression Analysis
  - Time Series Forecasting

Example: Predicting next quarter's sales based on historical trends.

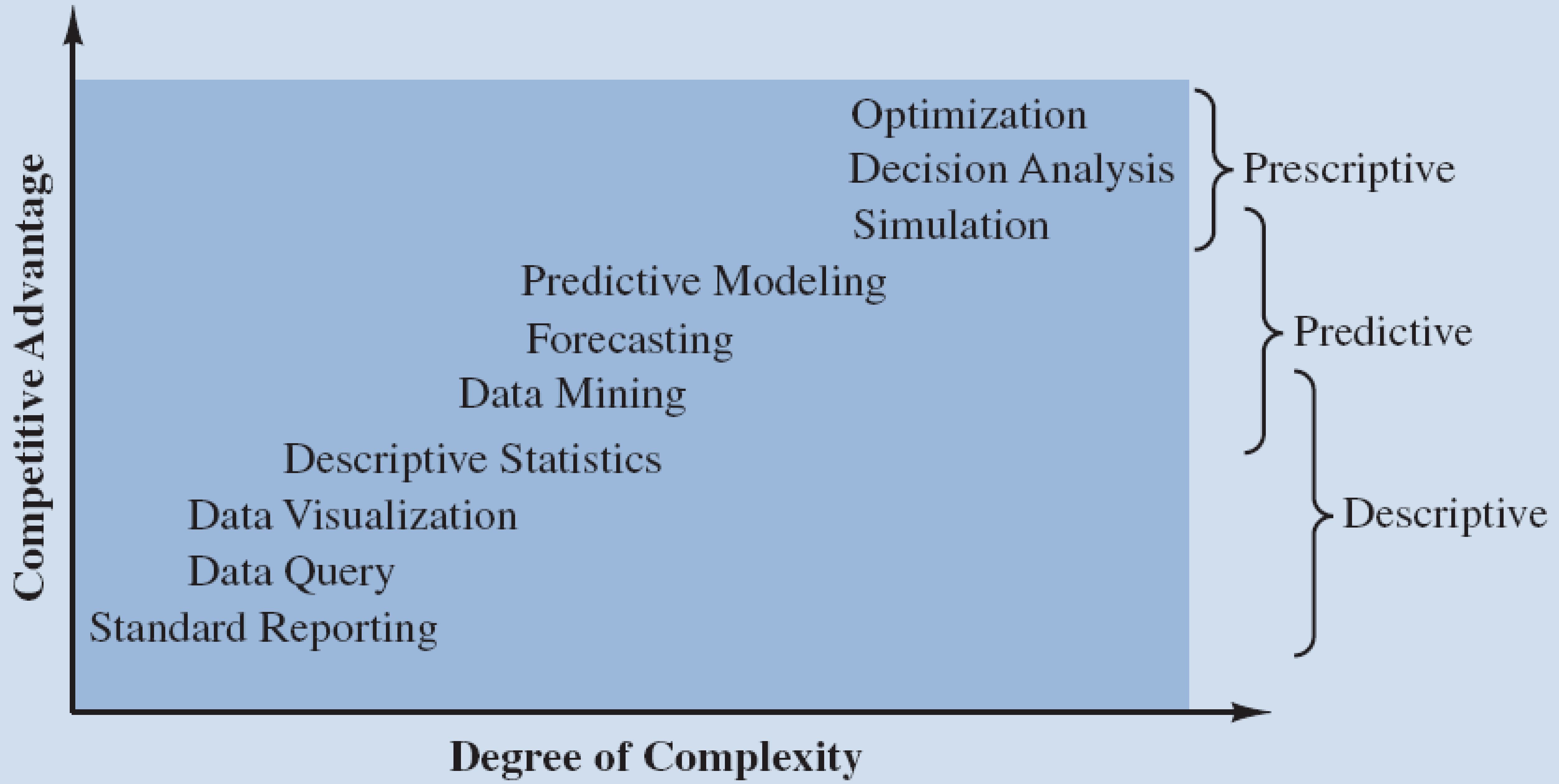
Categorization of Analytical Methods

(Types of Data Analytics)

Prescriptive Analytics

- Answers “**What should we do?**”
- Recommends **optimal actions** to achieve desired results.
- Methods & Tools
  - Optimization Algorithms
  - Simulation Modeling

Example: Suggesting the best marketing strategy to maximize customer engagement.

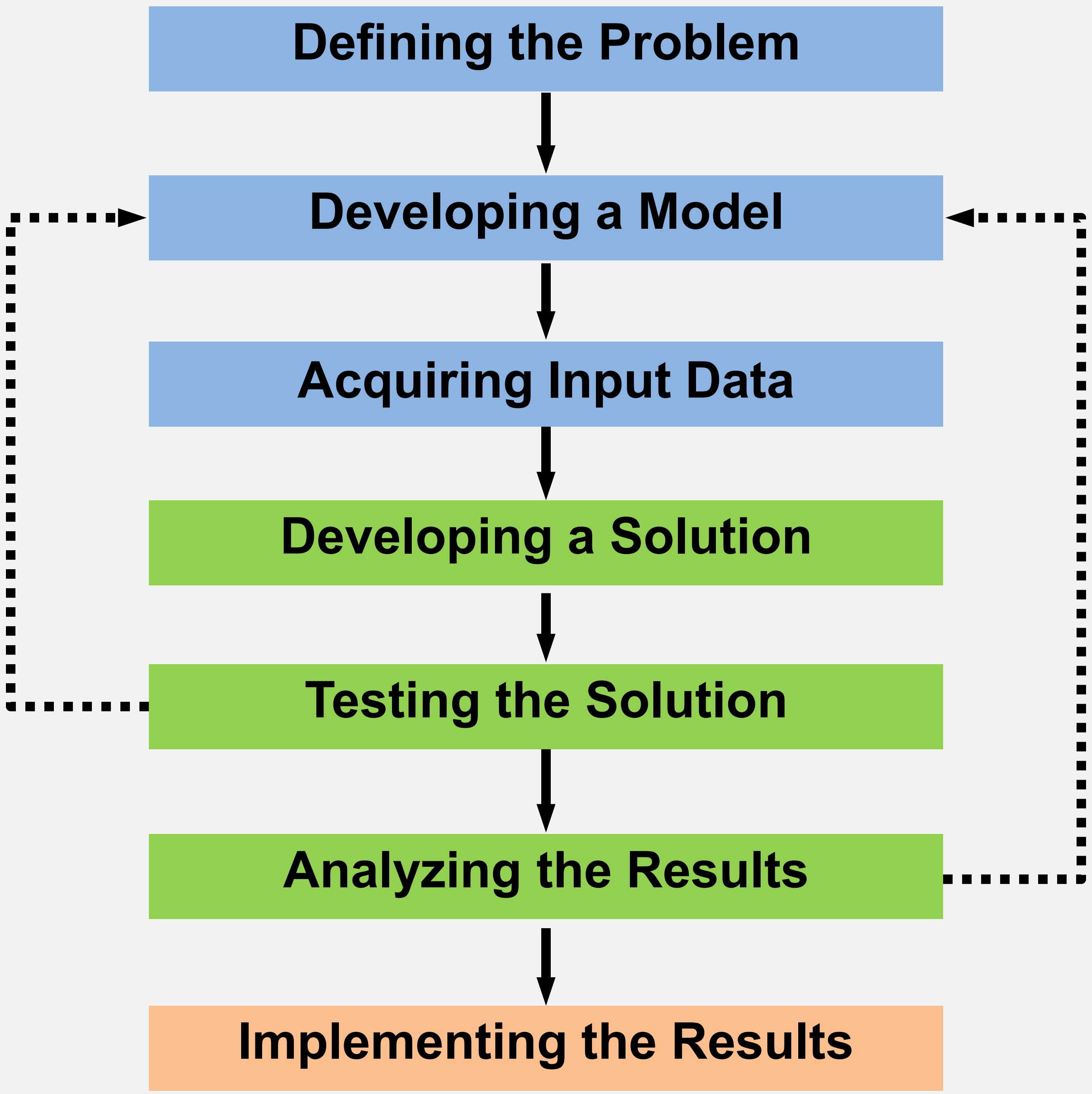


Source: Adapted from SAS.

Part III
The Quantitative Analysis Approach

The Quantitative Analysis Approach

The QA approach follows a structured sequence of steps as shown in the diagram.



Step 1: Defining the Problem

- ✓ This is the **most critical** and **most difficult** step.
- ✓ A **problem** is a **clear** and **concise statement** that gives **direction** and **meaning** to subsequent steps.
- ✓ To identify the problem, it must be:
  - **Clear** (Answers **Why**)
  - **Measurable** (Answers **What**)
  - **Specific** (Answers **How**)

Step 1: Defining the Problem

- ✓ Focus on the **root cause**, not just symptoms.
- ✓ It may be necessary to **concentrate** on only a **few of the problems** – selecting the **right problems** is very important.

Example:

Poor problem definition: *"Sales are low".*

Better problem definition: *"Monthly sales **decreased by 15%** due to **increased delivery time**".*

Step 1: Defining the Problem

Poor Problem Statement:

"How can we **reduce** student **drop-out rates**?"

Good Problem Statement:

"How can we **reduce** the student **drop-out rate** by **at least 10%** among **second-year college students** within **one academic year** through **targeted academic support** and **early intervention programs**?"

Step 2: Developing a Model

- ✓ A **model** is a **simplified representation** of a real-world **situation**.

- ✓ Models are **realistic**, **solvable**, and understandable **mathematical representations** of a situation.

- ✓ Models help us:
  - Understand complex systems
  - Predict outcomes
  - Compare alternatives

Step 2: Developing a Model

- ✓ Models generally consist of **variables** and **parameters**.
- ✓ **Variables** may be classified as **controllable** or **uncontrollable**.
  - **Controllable variables**, also called **decision variables**, represent **choices available** to the decision maker and are **typically unknown** until a solution is determined.

Step 2: Developing a Model

- **Uncontrollable variables** are environmental factors that cannot be **manipulated/changed** and are **random** and sometimes **unknown**.

- **Output variable** is the **outcome/result** computed by the model.

✓ **Parameters** are **known** and **fixed values** that define the environment of the model and **remain constant** during the analysis.

Step 2: Developing a Model

Example: Retail Store Operations

✓ **Controllable Variables (Decision Variables)**
- Number of units to produce or order
- Selling price of a product
- Number of employees scheduled per shift

✓ **Uncontrollable Variables (Randomness)**
- Customer demand
- Competitor pricing strategies
- Economic conditions (inflation, interest rates)
- Weather conditions affecting sales

Step 2: Developing a Model

Example: Retail Store Operations

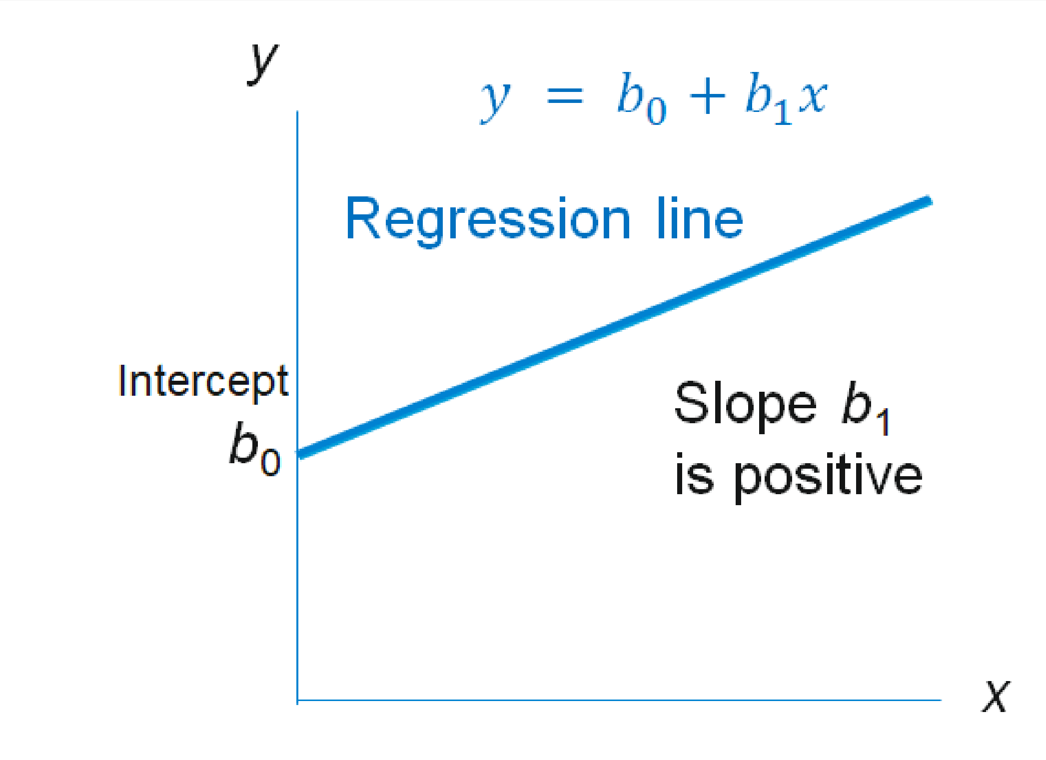
- ✓ **Output Variable(Outcome/Result)**
  - Profit
  - Store capacity utilization
  - Order fulfilment time

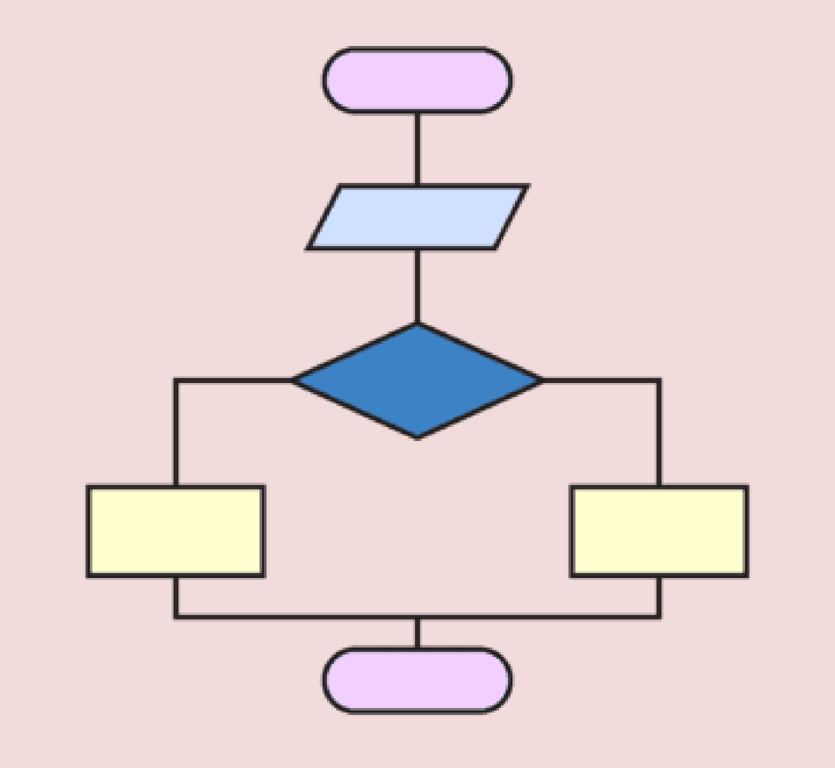
- ✓ **Parameters (Known and Fixed Inputs)**
  - Fixed costs (rent, salaries, insurance)
  - Variable cost per unit
  - Supplier contract prices
  - Tax rates and government fees

Step 2: Developing a Model

✓ **Types of Models**

- **Mathematical Models:** equations and formulas
- **Schematic Models**: flowcharts, diagrams
- **Physical Models:** scale models





Step 3: Acquiring Input Data

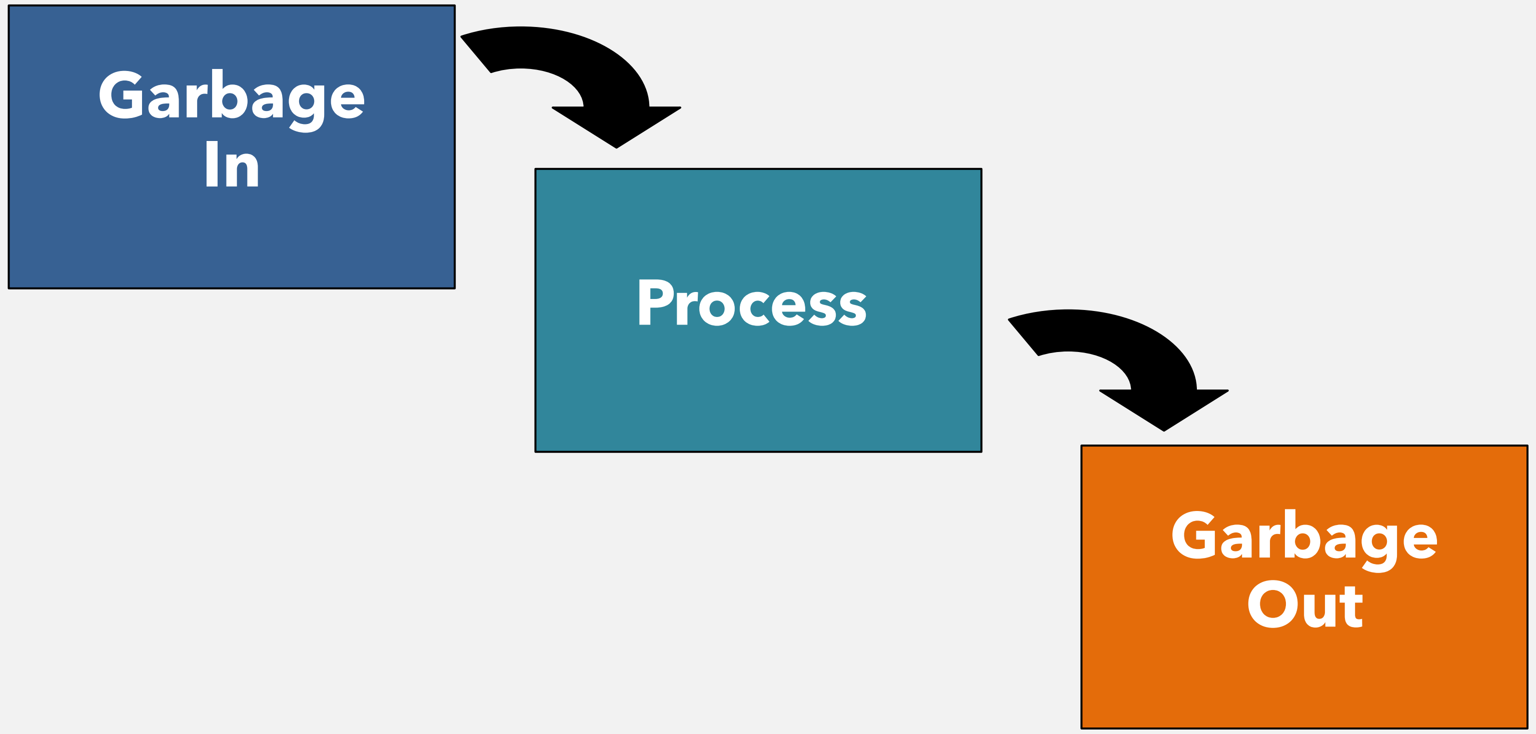
- ✓ **Data accuracy** is **critical**.
- ✓ Common **data sources**:
  - Company records
  - Surveys
  - Sensors and system logs
  - Historical data

Example: Collecting attendance and grade records.

Step 3: Acquiring Input Data

✓ **GIGO Principle:**

Garbage In, Garbage Out – poor data leads to poor decisions



Step 4: Developing a Solution

- ✓ In this step, the goal is to find the **optimal solution**.
- ✓ Common solution techniques:
  - Algebraic methods (use of mathematical equations)
  - Complete enumeration (listing)
  - Trial-and-error (brute force)
  - Algorithms (step-by-step procedure)

Example:
Trying different staffing levels to minimize cost while meeting service demand.

Step 5: Testing the Solution

- ✓ Both **input data** and the **model** should be **tested for accuracy** before analysis and implementation.
- ✓ **New data** can be collected to **test the model**.
- ✓ **Results** should be **logical**, **consistent**, and **represent** the **real situation** by asking :
  - Does the solution make sense?
  - Are results realistic?
  - What happens if inputs change?

Step 6: Analyzing the Results

- ✓ **Interpret the meaning** of the solution
- ✓ Determine the **implications** of the solution.
- ✓ Consider **organizational impact**.
- ✓ Perform **Sensitivity Analysis**: Measures how **changes in inputs affect outputs.**

Step 7: Implementing the Results

- ✓ Most QA **failures happen here**.
- ✓ Challenges include:
  - Resistance to change
  - Lack of management support
  - Misunderstanding of results

Key Ideas

- A **good solution** is **useless** if it is **not implemented properly**.
- **Changes occur over time**, so even successful implementations **must be monitored** to determine if modifications are necessary.

Advantages of Mathematical Modeling

Help decision makers formulate problems	Provides insights and information
Decision making	Save time and money

Deterministic vs Probabilistic Models

Deterministic Models
All values known with certainty
Example: payroll calculation

Probabilistic Models
Values based on probability
Example: demand forecasting

PART IV
Issues and Limitations in Quantitative Analysis

Common Issues in Quantitative Analysis

Problem Definition Issues

- Wrong problem identified
- Conflicting objectives
- Bias in assumptions

Model Development Issues

- Overly complex models
- Oversimplified assumptions
- Misalignment with real-world behavior

Common Issues in Quantitative Analysis

Data Issues

- Incomplete data
- Outdated data
- Inaccurate data

Solution and Implementation Issues

- Solutions difficult to understand
- Resistance from users
- Lack of organizational support

Key Limitations of Quantitative Analysis

- QA cannot fully capture **human behavior**
- QA relies heavily on **assumptions**
- Results are only as good as the **data**

Applications of Data Analytics

Customer Segmentation and Targeting

- Identify **distinct groups** of customers based on behavior, preferences, and demographics.

- Helps in designing **personalized marketing** strategies to improve engagement and sales.

Example: An e-commerce store grouping customers into “bargain hunters” and “premium buyers.”

Applications of Data Analytics

Demand Forecasting and Inventory Optimization

- Predict **future product demand** to ensure optimal stock levels.

- Reduces overstock and stockouts, improving **operational efficiency.**

Example: A retail chain forecasting seasonal demand for holiday products.

Applications of Data Analytics

Fraud Detection and Risk Management

- Detect **suspicious transactions** and prevent financial losses.
- Protects businesses from **fraud**, **theft**, and **data breaches**.

Example: A bank using analytics to detect unusual credit card spending patterns.

Applications of Data Analytics

Operational Efficiency Improvement

- Analyze internal processes to **identify bottlenecks** and **inefficiencies**.
- Lowers costs and **speeds up operations**.

Example: A manufacturing plant optimizing production scheduling to reduce downtime.

Applications of Data Analytics

Marketing Campaign Performance Analysis

- Evaluate the **effectiveness of marketing campaigns** across channels.
- Helps **allocate budget** to the most successful campaigns.

Example: Measuring ROI of Facebook ads versus email marketing campaigns.

Applications of Data Analytics

Product and Service Development

- Use analytics to identify **customer needs** and improve offerings.
- Increases **customer satisfaction** and **competitiveness.**

Example: Streaming platforms analyzing viewing data to decide what content to produce next.

END OF LESSON